

Technology-Enhanced Learning

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Intended Learning Outcomes

By the end of this session, you will be able to:

- Describe the CityUHK Digital Learning Ecosystem
- Apply generative AI for teaching, learning and assessments
- Use CityUHK Canvas for teaching, learning and assessments
- Describe VR and other DL tools

e-Learning is ...

Canvas? WiFi? (BC: Before COVID/ChatGPT)

Zoom? ChatGPT? (AC: After COVID/ChatGPT)

(e-)Learning is evolving...

Goal: More **Quality Time**
Through Technologies

Less Lecturing
More Active Learning

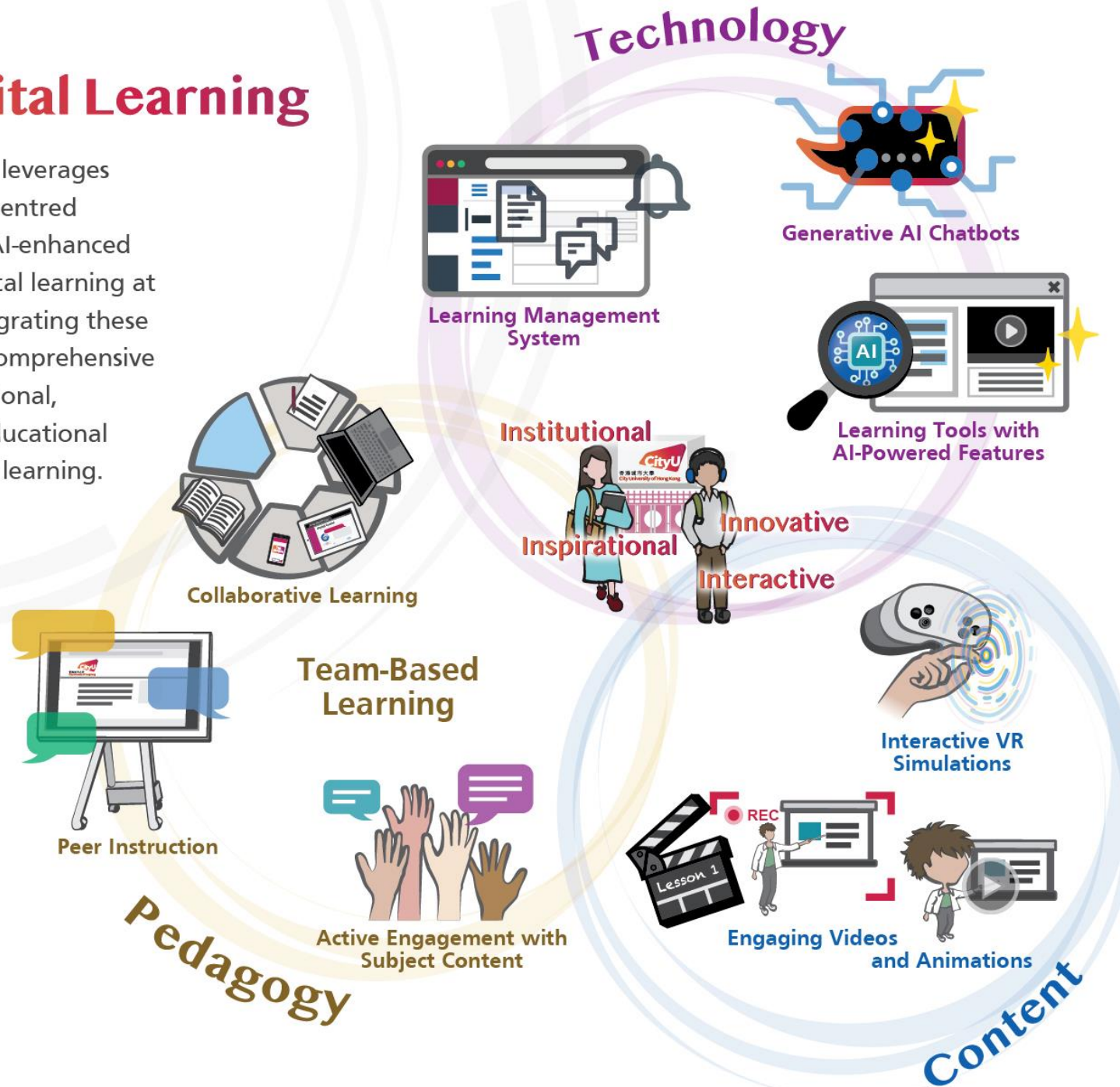
Digital Bloom's Taxonomy

Bloom's Taxonomy	Activities with Digital Tools	
Creating	 Programming  Animating  Filming  Podcasting  Wiki-ing	 Higher Order Thinking Skills Lower Order Thinking Skills
Evaluating	 Blog reviewing & commenting  Posting  Testing	
Analyzing	 Mind Mapping  Linking  Reverse Engineering	
Applying	 Editing  Calculating	
Understanding	 Categorizing  Tagging  Annotating	
Remembering	 Bookmarking  Bullet pointing  Highlighting	

Bloom's Digital Taxonomy by Sharon Keung is licensed under [CC BY-SA 4.0](#), based on the adoption of [Bloom's Digital Taxonomy](#) by Fractus Learning licensed under [CC BY-SA 4.0](#)

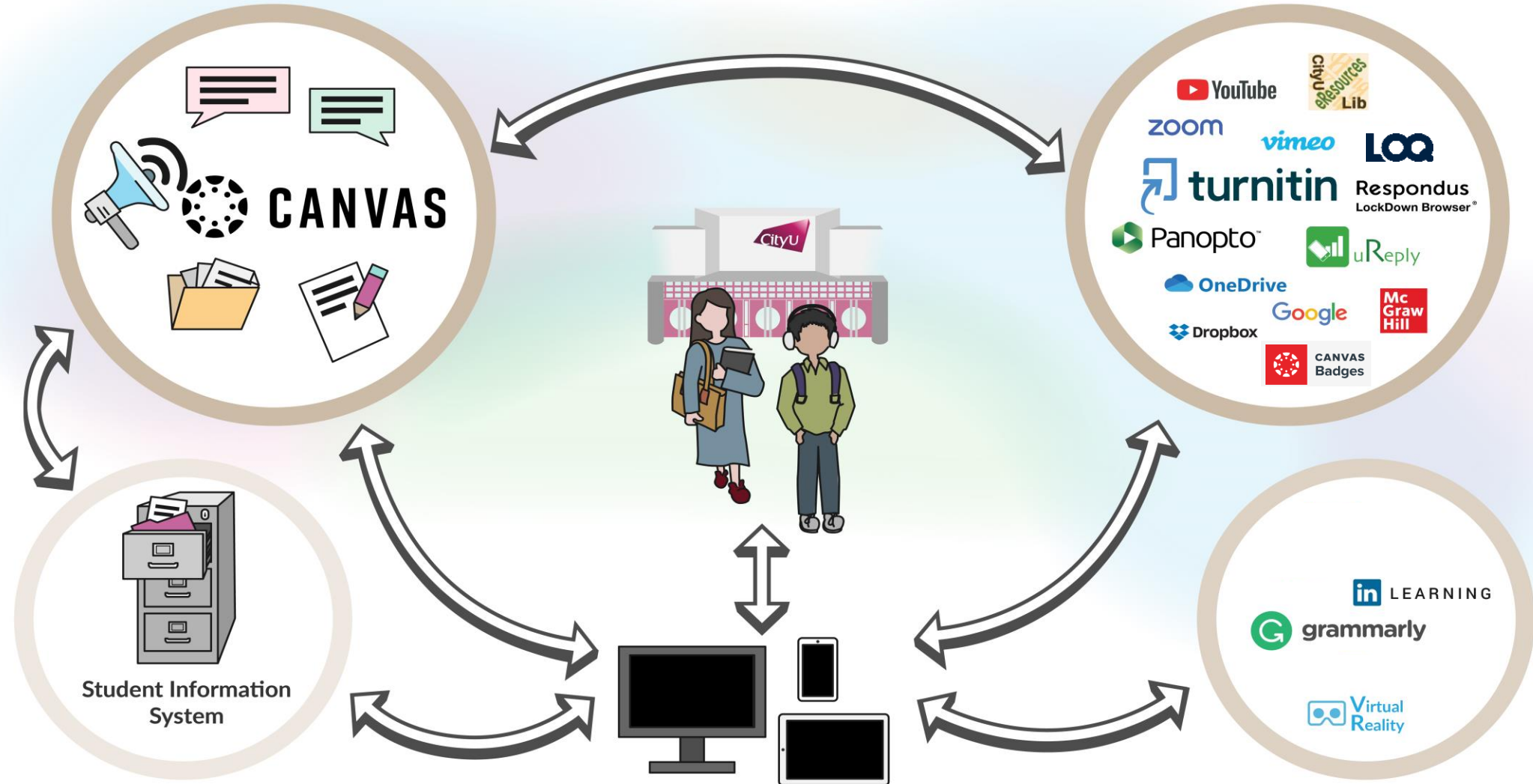
CityUHK Digital Learning

City University of Hong Kong leverages content innovation, student-centred interactive pedagogies, and AI-enhanced technologies to facilitate digital learning at the institutional level. By integrating these elements, students access a comprehensive environment with an inspirational, interactive, and innovative educational experience, fostering lifelong learning.



CityUHK Digital Learning Ecosystem

Illustrated by Betsy Wong



Scenario: Digital Team-Based Learning in CityUHK

Teachers are adopting team-based learning (TBL) to ensure active participation in classes and help students apply knowledge to real-world scenarios

Blending “spaces”:
Physical (classrooms)
Virtual (chatbot, VR)



Digital Learning Support in CityUHK

- For students (<https://www.cityu.edu.hk/dlsupport/student.html>)
 - FAQ <https://www.cityu.edu.hk/dlsupport/student-canvas-faq.html>
- For teachers (<https://www.cityu.edu.hk/dlsupport/teacher.html>)
 - FAQ <https://www.cityu.edu.hk/dlsupport/teacher-canvas-faq.html>
- Available DL tools (<https://www.cityu.edu.hk/dlsupport/tools.html>)
- Generative AI (<https://www.cityu.edu.hk/GenAI/>)
- Team-based learning (https://www.cityu.edu.hk/ted/dla/tbl_design.htm)

Application of AI in Teaching & Learning in Higher Education

Course Chatbot for 24/7 Authentic Learning

Using chatbots

- As a tutor
(feedback, guide, explain)
- As a study buddy
(brainstorm, reflect, argue)

Examples

- VCS: Students took on the role of veterinarians, interacting with the horse owner and making diagnoses

Team-based Learning (VM4115 / LI-4109)



[NEW TOPIC](#) [DISCLAIMER](#)

[ENGLISH](#) [廣東話](#) [普通話](#)



Can you tell me about the case?

Certainly! Here's the case:

A 6-month-old Thoroughbred foal named "Pidgeot" is presented with swelling in both front legs, particularly around the knees. The owner reports that Pidgeot has recently shown stiffness in his gait, especially when first getting up. The swelling has increased over the past two weeks. The foal has grown faster than the rest of the foals at the farm.

Scan here for a copy of the answer:



Course Chatbot for 24/7 Authentic Learning

- SLW: Guide learners analyze the court case and discuss the proposed sentiment [Intensively adopted]
- LT: Generate high-quality questions based on reading materials (through API)

Team-based Learning (SLW Demo)



NEW TOPIC DISCLAIMER

ENGLISH 廣東話 普通話

What kind of sentence would you suggest for this case? Give me a short answer.

Given the circumstances of the case, including the severity of the injury inflicted on Tang Wai-kit, the use of a weapon (chopper), and the fact that it was a domestic violence incident, a custodial sentence is warranted. However, considering mitigating factors such as Wong Siu-ling's lack of prior serious criminal record, her claims of provocation and self-defense, and her difficult marital situation, a starting point for sentencing could be around 8 to 10 years imprisonment. This would then be adjusted based on any further mitigating or aggravating factors presented during sentencing.

Team-based Learning (SLW Demo)



NEW TOPIC DISCLAIMER

ENGLISH 廣東話 普通話

hi, how would you impose on a defendant

To determine an appropriate sentence for Wong Siu-ling, who has been convicted of wounding with intent to cause grievous bodily harm under section 17 of the Offences Against the Person Ordinance (Cap.212), we need to consider several factors including sentencing principles, aggravating and mitigating factors, and relevant case law.

Sentencing Principles

1. ****Just Deserts or Retribution****: The punishment should be proportionate to the crime committed.
2. ****Deterrence****: Both specific deterrence (to prevent Wong from re-offending) and general deterrence (to discourage others from committing similar offenses).
3. ****Rehabilitation****: Assess whether there is potential for rehabilitation.
4. ****Incapacitation/Public Protection****: Consider if Wong poses a

Design extensive assessments

- Generate grading rubrics, evaluate the essay and provide feedback
- Generate low-stake MCQs for knowledge check

Create a set of 15 multiple-choice questions for the subject of [subject/course]. The questions should be tailored for [audience/age group/education level].

They must be structured according to Bloom's Taxonomy, with the difficulty distribution as follows: 1/3 of the questions should be of moderate difficulty, 1/3 should focus on application, and the remaining 1/3 should be at the evaluation level.

Each question should cover [specific concepts or topics]. Provide four answer choices for each question, clearly indicating the correct answer and providing explanations for the choice. Include plausible distractors that are relevant but incorrect.

Various Types of Instructional Videos for Instructions

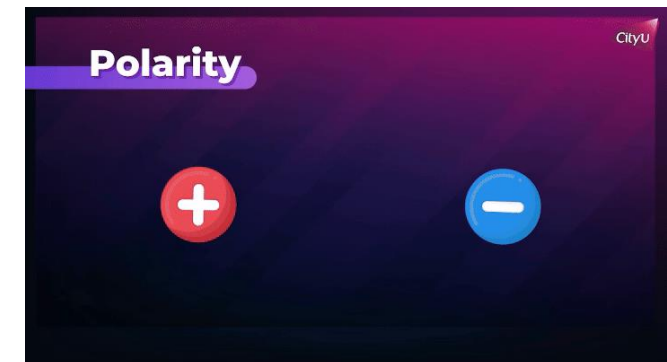
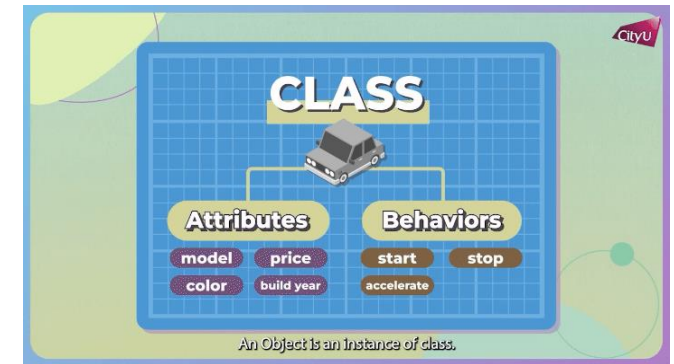
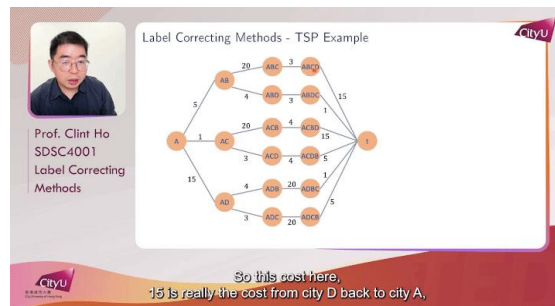


AI-Supported Videos

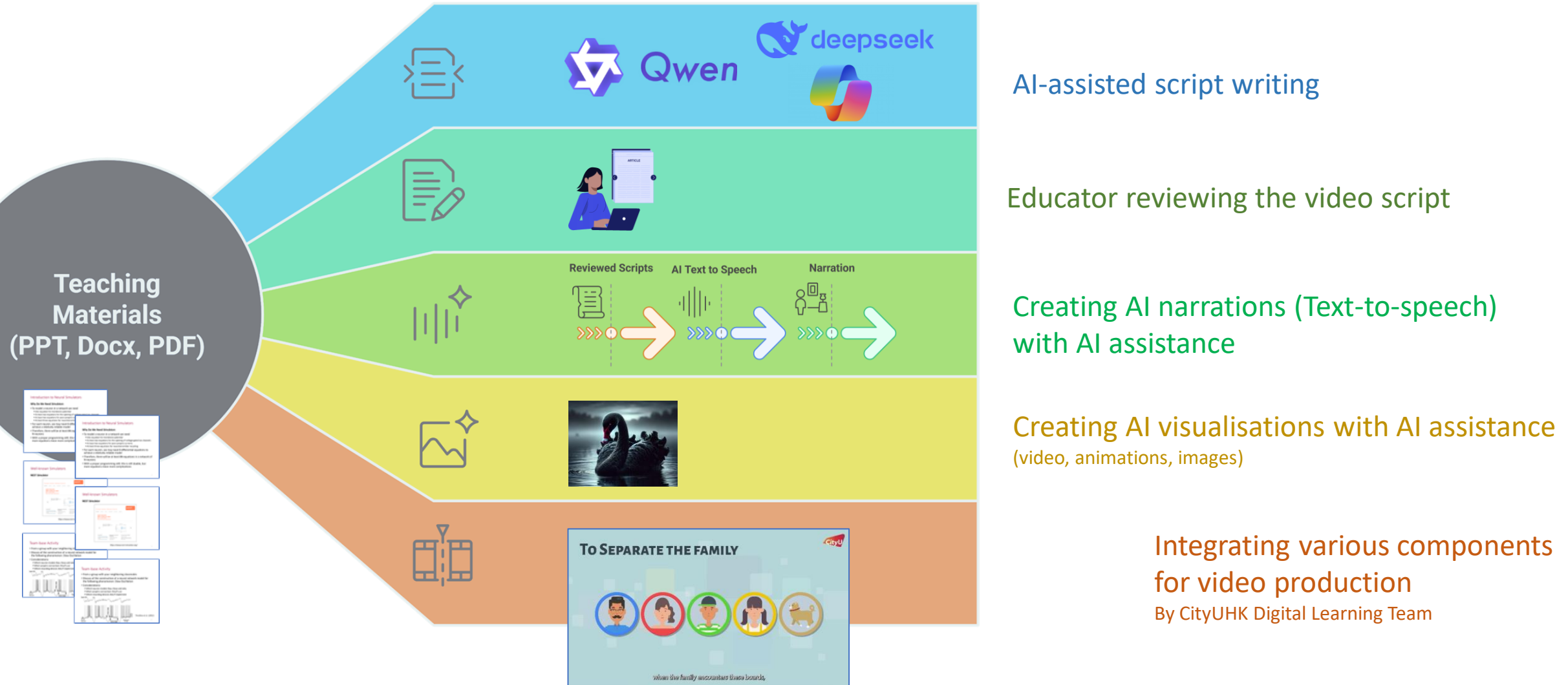
Green Screen Videos

Screen Recording

Animated Videos



AI-assisted Video Production in CityUHK



Hands-On

- Interact with the VCA horse diagnoses chatbot
 - Get into the system: <https://tbl.its.cityu.edu.hk/>
 - Search the room “SG8001 – Vet”
 - Interact with the bot, such as “Hello”
- Experiment with T&L prompts from Harvard or CityUHK
 - <https://github.com/motazsaad/Harvard-System-Prompt-Library>
 - <https://github.com/cityuhk-dl/DL-Prompt-Library/>
- Experiment with our VR headset

CityUHK Canvas

Canvas introduction video <http://go.cityu.hk/cp136j>

Key Functions in Canvas

- Add files
- Post an Announcement
- Design a Quiz
- Design and mark an Assignment
 - Turnitin Similarity and AI detection

Providing Feedback through SpeedGrader

- Preconfigured rubrics
- Comment Library
 - Stores and reuses common comments when grading assignments
- Peer review
 - Fosters interactive learning experiences
 - Enables students to reflect on their understanding of learned concepts, develop a deeper appreciation for different perspectives

Effective Communication Through Canvas

- Discussions
 - "focused discussions" to facilitate short-term exchanges
 - Sharing resources, collecting feedback from a one-off class activity
 - "threaded discussions" for in-depth exploration of complex problems or collaborative idea development
 - Stimulate student participation by grading discussions or requiring replies before they can view others' comments
- Announcements
- Inbox Messaging
- Comments on Assignments

How does Turnitin AI detection work?

Plagiarism:
Compare with
**published
records** in the
database

AI:
Calculate the
likelihood of texts
generated by
GenAI

How does it work?

0.30	This was written by me, a person.
0.42	Also this, penned by my hand.
0.68	Here starts the voice of the machine.
0.83	This is what the machine wrote.
0.97	What surprising predictability!
0.80	This writing lacks flavour.
0.61	Now back again to my own hand.
0.25	My words are delightfully spicy!
0.00	Thank you for reading my document.

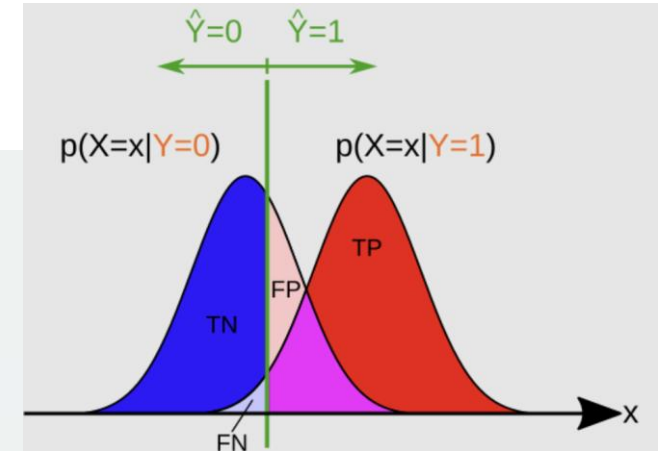
- Segments are analysed and assigned an AI writing prediction score
- Prediction scores are averaged and assigned per sentence
- Only sentences with a high prediction score are highlighted in the AI writing report



Interpreting the Turnitin AI detection score

"The risk of false positives from Turnitin's AI writing indicator is less than 1% for a document with over 20% likely AI-generated content."

Interpreting the score



--%

Submission does not meet minimum requirements.

0%

No Gen-AI content detected in the submission.

*%

1%-19% Gen-AI content detected in the submission.

25%

20%-100% Gen-AI content detected in the submission.

[Sharpr for svg version. original work by kakau in a png, CC BY-SA 3.0](#), via Wikimedia Commons

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Reference: <https://www.turnitin.com/blog/ai-writing-detection-what-academic-leaders-need-to-know-as-technology-matures>

Interpreting the Turnitin AI detection report



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FULL AI

You may use AI extensively throughout your work either as you wish, or as specifically directed in your assessment. Focus on directing AI to achieve your goals while demonstrating your critical thinking

Similarity 36%FlagsAI Writing 67%

comes from the word "bulaklak," which means "flower," as the island is said to have been covered in wildflowers in the past. A third theory suggests that the name Boracay is derived from the Spanish word "borracha," which means "inebriated." According to this theory, Spanish explorers gave the island this name because the local people were fond of drinking a fermented coconut wine called "tuba." Regardless of the origin of its name, Boracay is known around the world for its stunning beaches, crystal-clear waters, and vibrant nightlife. The island is also known for its nightlife. Boracay's bars and nightclubs offer a wide range of entertainment, from live music to fire dancers. The atmosphere is lively and vibrant, making it a perfect place to unwind after a long day at the beach. According to the Lonely Planet, "the nightlife here is a major draw, with a good mix of laid-back beach bars, swanky nightclubs, and live-music venues."

The island's other noteworthy beaches include Puka Beach and Bulabog Beach in addition to White Beach. Bulabog Beach is the preferred location for water sport enthusiasts, particularly windsurfers and kiteboarders, while Puka Beach is well-known

67% detected as AI ⓘ

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Submission Breakdown



1 AI-generated only	56%
Likely AI-generated text from a large-language model.	
2 AI-generated text that was AI-paraphrased	11%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.	

Sample of AI Report with AI generated content

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Gathering additional evidence for verifying the originality of the student's assignment



Additional evidence

Text manipulations	Formatting	Language	Content
• Hidden text • Replaced characters	• Inconsistent layout • Inconsistent fonts	• Unusual word choice or phrasing • Changes in spelling and/or grammar • Mixture of UK/US spelling • Inclusion of prompts from generative AI tools	• Lack of quotes or evidence • Invalid sources or citations that do not match with the reference section. • Inconsistent referencing style • Hallucinations (false information)

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We can also check the consistency of their writing log (Google Doc, MS Word 365), outlines, drafts, or previous essays

Ask students to prepare in advance

Creating the course AI policy

- Type of policy
- Conditions and contexts
 - Only for specific tasks or tools
 - Only when usage logs are provided
- Process and Consequences
- Declaration ([Case from PolyU](#))

Designing assessment tasks that make the use of AI less relevant

- Assess students holistically
 - Multiple times (i.e., more (in-class) assignments, continuous assessment)
 - Multiple modalities beyond text (e.g., video, infographic, vlog for CA)
- Asking questions that are challenges to AI
 - Specificity, Domain Knowledge, Real-world situation (Less online data available)
 - Complexity, Problem-Solving, Creativity (Out of the box)
 - Requires student's personal context (No digital data available)
- Make the assessment process visible to teachers
 - Explain the process through a reflection
 - Collaborative assessment and peer evaluation of other's process

Integrating AI into assignments

- Select a case from Harvard AI Pedagogy Project
 - <https://aipedagogy.org/assignments/>
 - <https://aipedagogy.org/assignment/interview-a-fictional-character/>
 - <https://aipedagogy.org/assignment/ai-validation/>
 - <https://aipedagogy.org/assignment/ai-sandwich/>
 - <https://aipedagogy.org/assignment/correct-a-bad-essay/>
- Many tasks ask students to grasp AI mechanics, biases, and limits, prompting reflection on societal impact, misinformation, and bias.
- Assignments emphasize collaboration—not replacement—encouraging creativity, problem-solving, and critical thinking. Students must scrutinize, justify, and compare AI outputs with human insights, using AI for scenario simulation.

Virtual Reality

Interactive Virtual Tours as Real-World Simulations

A fully immersive environment where students can interact with scenarios safely and without risk

- <https://www.cityu.edu.hk/ted/studentlan/dla/vr/>
 - Embedded videos and learning materials into the environment
- <https://www.cityu.edu.hk/smartclass/phylab-vr/>
- <https://www.cityu.edu.hk/smartclass/vrssp/>

